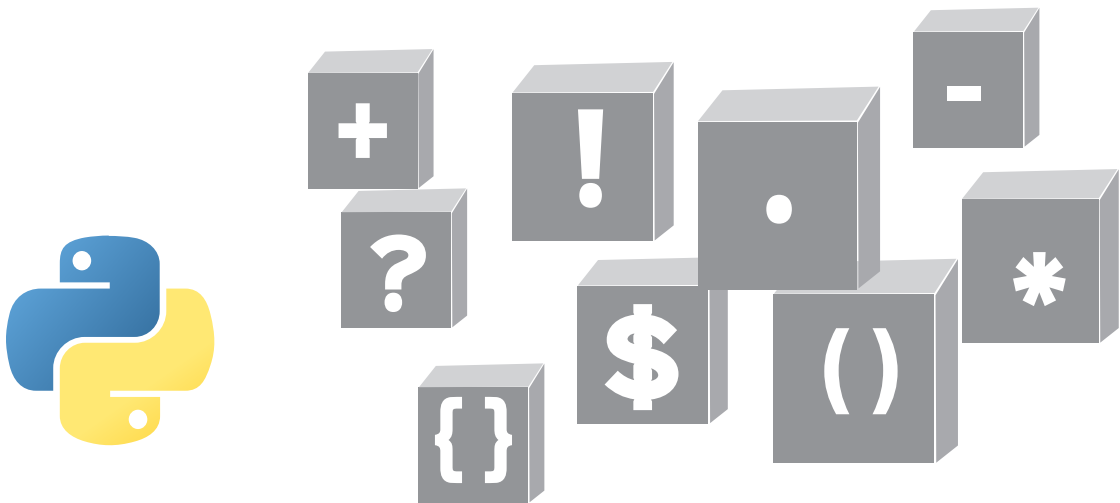


Working with Regular Expressions Using Python

Regular expressions contain a series of characters that define a pattern of text to be matched. Let's use Python for exploring the different things that regular expressions can do.



Regular expressions have become an integral part of natural language processing (NLP) and are very useful in the fields of text processing, information extraction and more. LLMs and machine learning models are being used today for tasks that can be easily done using regular expressions (regex). In fact, regex can even be used for many daily tasks in software development.

A regex is a special sequence of characters that is used to extract part of a string, or detect a particular string pattern. Log parsing is also a very important application of regular expressions. Here, we will use Python for exploring the different things that we can do using regex. We can even extract PII (personal identifiable information), emails, phone numbers, and more using regex.

Let us get started. Today the simplest way to run Python codes is to use Google Colab. This is useful for running any kind of Python projects including machine learning and data science models. Open <https://colab.research.google.com>, as shown in Figure 1.

In Python, the most popular library used for regex is 're'. It offers a set of functions that allow users to search, match, and manipulate strings based on specific patterns defined by regex. So, the first step before we move forward is to import this library using the following line of code:

```
import re
```

There are some special characters in the field of regular expressions, which help us perform different tasks. These are:

- \ - drop special meaning of char
- [] - represents a character class/set of characters
- ^ - Beginning of an expression
- \$ - Ending of an expression
- .
- Any character except the new line character ('\n')
- | - Means OR - Matches with characters that are separated by it
- ? - Matches zero or One occurrence (Verifies whether the given